

[MS-GPAC]:

Group Policy: Audit Configuration Extension

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1 Introduction

This document specifies the Group Policy: Audit Policy Configuration Protocol, which provides a mechanism for an administrator to control advanced audit policies on clients.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Active Directory: The Windows implementation of a general-purpose directory service, which uses **LDAP** as its primary access protocol. Active Directory stores information about a variety of objects in the network such as user accounts, computer accounts, groups, and all related credential information used by Kerberos [\[MS-KILE\]](#). Active Directory is either deployed as **Active Directory Domain Services (AD DS)** or Active Directory Lightweight Directory Services (AD LDS), which are both described in [\[MS-ADOD\]](#): Active Directory Protocols Overview.

Active Directory Domain Services (AD DS): A directory service (DS) implemented by a **domain controller (DC)**. The DS provides a data store for objects that is distributed across multiple **DCs**. The DCs interoperate as peers to ensure that a local change to an object replicates correctly across DCs. AD DS is a deployment of **Active Directory** [\[MS-ADTS\]](#).

Active Directory object: A set of directory objects that are used within **Active Directory** as defined in [\[MS-ADTS\]](#) section 3.1.1. An **Active Directory object** can be identified by a dsname. See also directory object.

Administrative tool: An implementation-specific tool, such as the Group Policy Management Console, that allows administrators to read and write **policy settings** from and to a **Group Policy Object (GPO)** and policy files. The Group Policy Administrative tool uses the Extension list of a GPO to determine which Administrative tool extensions are required to read settings from and write settings to the logical and physical components of a GPO.

advanced audit policy: The global audit policy settings pertaining to auditing as described in this specification.

attribute: A characteristic of some object or entity, typically encoded as a name/value pair.

audit policy: The global audit policy settings pertaining to auditing as described in [\[MS-GPSB\]](#) section 2.2.4.

Augmented Backus-Naur Form (ABNF): A modified version of Backus-Naur Form (BNF), commonly used by Internet specifications. ABNF notation balances compactness and simplicity with reasonable representational power. ABNF differs from standard BNF in its definitions and uses of naming rules, repetition, alternatives, order-independence, and value ranges. For more information, see [\[RFC5234\]](#).

client-side extension GUID (CSE GUID): A **GUID** that enables a specific client-side extension on the Group Policy client to be associated with policy data that is stored in the logical and physical components of a **Group Policy Object (GPO)** on the Group Policy server, for that particular extension.

computer-scoped Group Policy Object path: A scoped Group Policy Object (GPO) path that ends in "\\Machine".

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a **domain controller**

(**DC**) and host a member list that identifies all members of the domain, as well as optionally hosting the **Active Directory** service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

domain controller (DC): The service, running on a server, that implements **Active Directory**, or the server hosting this service. The service hosts the data store for objects and interoperates with other **DCs** to ensure that a local change to an object replicates correctly across all **DCs**. When **Active Directory** is operating as **Active Directory Domain Services (AD DS)**, the **DC** contains full NC replicas of the configuration naming context (config NC), schema naming context (schema NC), and one of the domain NCs in its forest. If the **AD DS DC** is a global catalog server (GC server), it contains partial NC replicas of the remaining domain NCs in its forest. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5.2 and [\[MS-ADTS\]](#). When **Active Directory** is operating as Active Directory Lightweight Directory Services (AD LDS), several AD LDS **DCs** can run on one server. When **Active Directory** is operating as **AD DS**, only one **AD DS DC** can run on one server. However, several AD LDS **DCs** can coexist with one **AD DS DC** on one server. The AD LDS **DC** contains full NC replicas of the config NC and the schema NC in its forest. The domain controller is the server side of Authentication Protocol Domain Support [\[MS-APDS\]](#).

globally unique identifier (GUID): A term used interchangeably with universally unique identifier (UUID) in Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the value. Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [\[C706\]](#) have to be used for generating the GUID. See also universally unique identifier (UUID).

Group Policy: A mechanism that allows the implementer to specify managed configurations for users and computers in an **Active Directory** service environment.

Group Policy Object (GPO): A collection of administrator-defined specifications of the policy settings that can be applied to groups of computers in a domain. Each GPO includes two elements: an object that resides in the **Active Directory** for the domain, and a corresponding file system subdirectory that resides on the sysvol DFS share of the Group Policy server for the domain.

Group Policy server: A server holding a database of **Group Policy Objects (GPOs)** that can be retrieved by other machines. The Group Policy server has to be a **domain controller (DC)**.

Lightweight Directory Access Protocol (LDAP): The primary access protocol for **Active Directory**. Lightweight Directory Access Protocol (LDAP) is an industry-standard protocol, established by the Internet Engineering Task Force (IETF), which allows users to query and update information in a directory service (DS), as described in [\[MS-ADTS\]](#). The Lightweight Directory Access Protocol can be either version 2 [\[RFC1777\]](#) or version 3 [\[RFC3377\]](#).

policy setting: A statement of the possible behaviors of an element of a domain member computer's behavior that can be configured by an administrator.

security identifier (SID): An identifier for security principals that is used to identify an account or a group. Conceptually, the **SID** is composed of an account authority portion (typically a **domain**) and a smaller integer representing an identity relative to the account authority, termed the relative identifier (RID). The **SID** format is specified in [\[MS-DTYP\]](#) section 2.4.2; a string representation of **SIDs** is specified in [\[MS-DTYP\]](#) section 2.4.2 and [\[MS-AZOD\]](#) section 1.1.1.2.

share: A resource offered by a Common Internet File System (CIFS) server for access by CIFS clients over the network. A share typically represents a directory tree and its included files (referred to commonly as a "disk share" or "file share") or a printer (a "print share"). If the information about the share is saved in persistent store (for example, Windows registry) and

reloaded when a file server is restarted, then the share is referred to as a "sticky share". Some share names are reserved for specific functions and are referred to as special shares: IPC\$, reserved for interprocess communication, ADMIN\$, reserved for remote administration, and A\$, B\$, C\$ (and other local disk names followed by a dollar sign), assigned to local disk devices.

system access control list (SACL): An access control list (ACL) that controls the generation of audit messages for attempts to access a securable object. The ability to get or set an object's **SACL** is controlled by a privilege typically held only by system administrators.

ticket-granting ticket (TGT): A special type of ticket that can be used to obtain other tickets. The TGT is obtained after the initial authentication in the Authentication Service (AS) exchange; thereafter, users do not need to present their credentials, but can use the TGT to obtain subsequent tickets.

token: A set of rights and privileges for a given user.

tool extension GUID or administrative plug-in GUID: A GUID defined separately for each of the user policy settings and computer policy settings that associates a specific administrative tool plug-in with a set of policy settings that can be stored in a **Group Policy Object (GPO)**.

Universal Naming Convention (UNC): A string format that specifies the location of a resource. For more information, see [MS-DTYP] section 2.2.57.

UTF-8: A byte-oriented standard for encoding Unicode characters, defined in the Unicode standard. Unless specified otherwise, this term refers to the UTF-8 encoding form specified in [UNICODE5.0.0/2007] section 3.9.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [RFC2119]. All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-GPOL] Microsoft Corporation, "[Group Policy: Core Protocol](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2251] Wahl, M., Howes, T., and Kille, S., "Lightweight Directory Access Protocol (v3)", RFC 2251, December 1997, <https://www.rfc-editor.org/info/rfc2251>

[RFC4234] Crocker, D., Ed., and Overell, P., "Augmented BNF for Syntax Specifications: ABNF", RFC 4234, October 2005, <https://www.rfc-editor.org/info/rfc4234>

1.2.2 Informative References

[MS-FASOD] Microsoft Corporation, "[File Access Services Protocols Overview](#)".

[MS-GPSB] Microsoft Corporation, "[Group Policy: Security Protocol Extension](#)".

[MSDN-SDDL] Microsoft Corporation, "Security Descriptor String Format", <http://msdn.microsoft.com/en-us/library/aa379570.aspx>

1.3 Overview

The Group Policy: Audit Configuration Extension to the Group Policy: Core Protocol [\[MS-GPOL\]](#) enables **advanced audit policies** to be distributed to multiple clients so that these clients can enforce the policies in accordance with the intentions of the administrator.

1.3.1 Background

The Group Policy: Core Protocol, as specified in [\[MS-GPOL\]](#), allows clients to discover and retrieve **policy settings** created by administrators of a **domain**. These settings are persisted within **Group Policy Objects (GPOs)** that are assigned to Policy Target accounts in **Active Directory**. Policy Target accounts are either computer accounts or user accounts in Active Directory. Each client uses **Lightweight Directory Access Protocol (LDAP)** to determine what GPOs are applicable to it by consulting the **Active Directory objects** corresponding to both its computer account and the user accounts of any users logging on to the client computer.

On each client, each GPO is interpreted and acted upon by client plug-ins. The client plug-ins that are responsible for a given GPO are specified using an attribute on the GPO. This attribute specifies a list of **globally unique identifier (GUID)** pairs. The first GUID of each pair is referred to as a **client-side extension GUID (CSE GUID)**. The second GUID of each pair is referred to as a **tool extension GUID**.

For each GPO that is applicable to a client, the client consults the CSE GUID listed in the GPO to determine what client plug-in on the client will handle the GPO. The client then invokes the client plug-in to handle the GPO.

A client plug-in uses the contents of the GPO to retrieve settings specific to the client plug-in in a manner specific to the client plug-in. After the client plug-in-specific settings are retrieved, the client plug-in uses those settings to perform the client plug-in-specific processing.

1.3.2 Audit Configuration Extension Overview

Advanced audit policies contain settings that enable the underlying audit subsystem to determine which activities to monitor and log in the security event log. Advanced audit policies contain three main types of settings:

- Audit subcategory settings
- Audit options
- Global object access policy

The following major steps are involved in advanced audit policy configuration:

1. Advanced audit policy authoring
2. Advanced audit policy assignment
3. Advanced audit policy distribution

Advanced audit policy authoring is enabled through an **administrative tool** for the Group Policy: Core Protocol specified in [\[MS-GPOL\]](#) with an administrative-side plug-in for behavior specific to this protocol. The plug-in allows an administrator to author advanced audit policies within an implementation-specific tool providing a graphical user interface. The plug-in then saves the advanced audit policies into files with a format specified in this document, and stores them on a file **share** that is accessible by file access protocol sequences as described in [\[MS-FASOD\]](#).

Advanced audit policy assignment is performed by the Group Policy: Core Protocol administrative tool, which constructs **GPOs**, as specified in [MS-GPOL] section 2.2.8.1. Each GPO contains a reference to the network path using the **Universal Naming Convention (UNC)** where the advanced audit policy files generated by the protocol administrative plug-in need to be fetched from using file access protocol sequences.

Advanced audit policy distribution involves a corresponding protocol-specific **Group Policy** plug-in on the client machine, which is invoked to process any GPO that refers to advanced audit policy settings. The advanced audit policy protocol client-side plug-in locates the advanced audit policy as specified in section [3.1.4](#), transfers the advanced audit policy files by using file access protocol sequences, and then uses the advanced audit policy files to configure the client's advanced audit policy, audit options, and global object access auditing settings.

1.3.2.1 Audit Subcategory Settings

The advanced audit policy allows administrators to select only the behaviors to monitor and to exclude audit results for behaviors that are of no concern to them or behaviors that create an excessive number of log entries. These settings are grouped into the following nine main audit categories containing the audit subcategories listed here:

System audit:

- Security State Change
- Security System Extension
- System Integrity
- IPsec Driver
- Other System Events

Logon/Logoff audit:

- Logon
- Logoff
- Account Lockout
- IPsec Main Mode
- Ipsec Quick Mode
- Ipsec Extended Mode
- Special Logon
- Other Logon/Logoff Events
- Network Policy Server
- User/Device Claims

Group Membership

Object access audit:

File system

Registry

Kernel Object

SAM

Certification Services

Application Generated

Handle Manipulation

File Share

Filtering Platform Packet Drop

Filtering Platform Connection

Other Object Access Events

Detailed File Share

Removable Storage

Central Access Policy Staging

Privilege use audit:

Sensitive Privilege Use

Non Sensitive Privilege Use

Other Privilege Use Events

Detailed tracking audit:

Process Creation

Process Termination

DPAPI Activity

RPC Events

PNP Activity

Policy change audit:

Audit Policy Change

Authentication Policy Change

Authorization Policy Change

MPSSVC Rule-Level Policy Change

Filtering Platform Policy Change

Other Policy Change Events

Account management audit:

User Account Management

Computer Account Management

Security Group Management

Distribution Group Management

Application Group Management

Other Account Management Events

Directory Service access audit:

Directory Service Access

Directory Service Changes

Directory Service Replication

Detailed Directory Service Replication

Account Logon audit:

Credential Validation

Kerberos Service Ticket Operations

Other Account Logon Events

Kerberos Authentication Service

For more information about audit subcategories, see section [2.2.1](#).

1.3.2.2 Audit Options

Audit options are settings that enable or disable functionality of the audit subsystem. These settings include crashing the system on audit failures, full privilege auditing, auditing of base objects, and auditing of base directories.

For more information about audit options, see section [2.2.2](#).

1.3.2.3 Global Object Access Policy

The global object access policy contains a set of system access control lists that are applied to whole resource managers like the File System and Registry.

For more information about global object access policy, see section [2.2.3](#).

1.4 Relationship to Other Protocols

This protocol depends on Group Policy: Core Protocol specified in [\[MS-GPOL\]](#) to provide a list of applicable GPOs. It also transmits Group Policy settings and instructions between the client and the **Group Policy server** by reading and writing files. See [\[MS-FASOD\]](#) for an overview of file access services concepts. The following diagram illustrates these relationships.

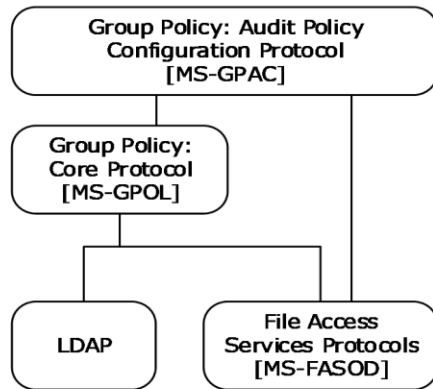


Figure 1: Group Policy: Audit Configuration Extension protocol relationship diagram

1.5 Prerequisites/Preconditions

The prerequisites for Group Policy: Audit Configuration Extension are the same as those for the Group Policy: Core Protocol [\[MS-GPOL\]](#).

1.6 Applicability Statement

Group Policy: Audit Configuration Extension is only applicable within the Group Policy framework.

1.7 Versioning and Capability Negotiation

None.

1.8 Vendor-Extensible Fields

None.

1.9 Standards Assignments

Group Policy: Audit Configuration Extension defines CSE GUID and tool extension GUID, as specified in [\[MS-GPOL\]](#) (section 1.8). The following table shows the assignments.

Parameter	Value
CSE GUID	{f3ccc681-b74c-4060-9f26-cd84525dca2a}
Tool extension GUID (computer policy settings)	{0F3F3735-573D-9804-99E4-AB2A69BA5FD4}

2 Messages

2.1 Transport

The Group Policy: Audit Configuration Extension requires file access as specified for use in the Group Policy: Core Protocol [\[MS-GPOL\]](#). All messages MUST be exchanged over file access protocols between the client and server, as specified in section [2.2](#).

The Group Policy: Core Protocol uses this protocol's **CSE GUID** and **tool extension GUID** values to invoke this protocol only to access **GPOs** that require processing by this protocol.

2.2 Message Syntax

Messages exchanged in the Group Policy: Audit Configuration Extension correspond to advanced audit policy files transferred by using file access protocol sequences as described in [\[MS-FASOD\]](#). The protocol is driven through the exchange of these messages, as specified in section [3](#).

All advanced audit policy files processed by the Group Policy: Audit Configuration Extension are **UTF-8** encoded and based on the following file syntax.

```
CSVFile = Header SubcategorySettings AuditOptions GlobalObjectAccessAuditSettings
Header = "Machine Name,Policy Target,Subcategory,Subcategory GUID,Inclusion Setting,Exclusion
Setting,Setting Value" LineBreak
```

The preceding syntax is given in the **Augmented Backus-Naur Form (ABNF)** grammar, as specified in [\[RFC4234\]](#) and as augmented by the following rules.

```
LineBreak = CRLF
String = *(ALPHANUM / %d47 / %d45 / %d58 / %d59)
StringWithSpaces = String / String Wsp StringWithSpaces
QuotedString = DQUOTE *(%x20-21 / %x23-7E) DQUOTE
Wsp = *WSP
ALPHANUM = ALPHA / DIGIT
GUID
    = %x7B time-low hyphen time-mid hyphen
      time-high-and-version hyphen
      clock-seq-and reserved
      clock-seq-low hyphen node %x7D
time-low
    = hexOctet hexOctet hexOctet hexOctet
time-mid
    = hexOctet hexOctet
time-high-and-version
    = hexOctet hexOctet
clock-seq-and-reserved
    = hexOctet
clock-seq-low
    = hexOctet
node
    = hexOctet hexOctet hexOctet
      hexOctet hexOctet hexOctet
hexOctet
    = hexDigit hexDigit
hexDigit
    = digit / a / b / c / d / e / f
digit
    = "0" / "1" / "2" / "3" / "4" / "5" / "6" / "7" /
      "8" / "9"
hyphen
    = "-"
a
    = "a" / "A"
b
    = "b" / "B"
c
    = "c" / "C"
d
    = "d" / "D"
e
    = "e" / "E"
f
    = "f" / "F"
```

2.2.1 Subcategory Settings

This section defines settings that enable an administrator to set the subcategory settings for an advanced audit policy. The syntax for the entries in this category **MUST** be as follows.

```
SubcategorySettings = SubcategorySetting / SubcategorySetting / SubcategorySetting
SubcategorySetting = MachineName "," PolicyTarget "," Subcategory "," SubcategoryGUID ","
InclusionSetting "," ExclusionSetting "," SettingValue LineBreak
```

2.2.1.1 Policy Target

This section defines the possible values for the PolicyTarget **attribute**, which enables an administrator to specify whether to set the audit subcategory for a system advanced audit policy or a specific user. The syntax for the entries in this category **MUST** be as follows.

```
PolicyTarget = "System" / UserSID
```

The value of PolicyTarget **MUST** be one of the following:

1. A value of "System": Indicates that this is a system audit subcategory setting.
2. A UserSID: Indicates that this is a per-user audit subcategory setting.

UserSID is the string representation of the **security identifier (SID)** of a user account. The syntax for the entries in this category **MUST** be as follows.

```
UserSID = String
```

The UserSID string **MUST** use the standard S-R-I-S-S... format for SID strings, as specified in [\[MS-DTYP\]](#) (section 2.4.2).[<1>](#)

2.2.1.2 Subcategory and SubcategoryGUID

This section defines how the Subcategory and SubcategoryGUID values are used by the audit configuration client-side plug-in.

The Subcategory field is for user reference only and is ignored when the advanced audit policy is applied by the audit configuration client-side plug-in.

The syntax for the entries in this category **MUST** be as follows.

```
Subcategory = StringWithSpaces / QuotedString
SubcategoryGUID = GUID
```

The SubcategoryGUID allows administrators to identify audit subcategories to enable or disable in the client's system or per-user advanced audit policy. For more information about enabling or disabling audit subcategories, see section [2.2.1.3](#).

The following table provides an explanation for the valid **SubcategoryGUID** values.

SubcategoryGUID	Purpose
{0CCE9213-69AE-11D9-	Identifies the IPsec Driver audit subcategory.

SubcategoryGUID	Purpose
BED3-505054503030}	This subcategory audits events that are generated by the IPsec filter driver.
{0CCE9212-69AE-11D9-BED3-505054503030}	Identifies the System Integrity audit subcategory. This subcategory audits events that violate the integrity of the security subsystem.
{0CCE9211-69AE-11D9-BED3-505054503030}	Identifies the Security System Extension audit subcategory. This subcategory audits events related to security system extensions or services.
{0CCE9210-69AE-11D9-BED3-505054503030}	Identifies the Security State Change audit subcategory. This subcategory audits events generated by changes in the security state of the computer.
{0CCE9214-69AE-11D9-BED3-505054503030}	Identifies the Other System Events audit subcategory. This subcategory SHOULD audit the following event: Cryptography key file and migration operations.
{0CCE9243-69AE-11D9-BED3-505054503030}	Identifies the Network Policy Server audit subcategory. This subcategory audits events generated by RADIUS (IAS) and Network Access Protection (NAP) user access requests. These requests can be Grant, Deny, Discard, Quarantine, Lock, and Unlock.
{0CCE921C-69AE-11D9-BED3-505054503030}	Identifies the Other Logon/Logoff Events audit subcategory. This subcategory audits other events related to logon and logoff that are not included in the Logon/Logoff category.
{0CCE921B-69AE-11D9-BED3-505054503030}	Identifies the Special Logon audit subcategory. This subcategory audits events generated by special logons.
{0CCE921A-69AE-11D9-BED3-505054503030}	Identifies the IPsec Extended Mode audit subcategory. This subcategory audits events generated by Internet Key Exchange protocol (IKE) and Authenticated Internet Protocol (AuthIP) during Extended Mode negotiations.
{0CCE9219-69AE-11D9-BED3-505054503030}	Identifies the IPsec Quick Mode audit subcategory. This subcategory audits events generated by Internet Key Exchange protocol (IKE) and Authenticated Internet Protocol (AuthIP) during Quick Mode negotiations.
{0CCE9218-69AE-11D9-BED3-505054503030}	Identifies the IPsec Main Mode audit subcategory. This subcategory audits events generated by Internet Key Exchange protocol (IKE) and Authenticated Internet Protocol (AuthIP) during Main Mode negotiations.
{0CCE9217-69AE-11D9-BED3-505054503030}	Identifies the Account Lockout audit subcategory. This subcategory audits events generated by a failed attempt to log on to an account that is locked out.
{0CCE9216-69AE-11D9-BED3-505054503030}	Identifies the Logoff audit subcategory. This subcategory audits events generated by closing a logon session. These events occur on the computer that was accessed. For an interactive logon, the security audit event is generated on the computer that the user account logged on to.
{0CCE9215-69AE-11D9-BED3-505054503030}	Identifies the Logon audit subcategory. This subcategory audits events generated by user account logon attempts on a computer.
{0CCE9223-69AE-11D9-BED3-505054503030}	Identifies the Handle Manipulation audit subcategory. This subcategory audits events generated when a handle to an object is opened or closed. Only objects with a matching SACL generate security audit events. Open and close handle events will be audited when both the Handle Manipulation subcategory is enabled along with the corresponding resource manager identified by other Object Access audit subcategory, like File System or Registry. Enabling Handle Manipulation causes implementation-specific security event data to be logged identifying the

SubcategoryGUID	Purpose
	permissions that were used to grant or deny the access requested by the user; this is also known as "Reason for access".
{0CCE9244-69AE-11D9-BED3-505054503030}	Identifies the Detailed File Share audit subcategory. This subcategory audits every attempt to access objects in a shared folder.
{0CCE9227-69AE-11D9-BED3-505054503030}	Identifies the Other Object Access Events audit subcategory. This subcategory audits events generated by the management of Task Scheduler jobs or COM+ objects.
{0CCE9226-69AE-11D9-BED3-505054503030}	Identifies the Filtering Platform Connection audit subcategory. This subcategory audits connections that are allowed or blocked by WFP.
{0CCE9225-69AE-11D9-BED3-505054503030}	Identifies the Filtering Platform Packet Drop audit subcategory. This subcategory audits packets that are dropped by Windows Filtering Platform (WFP).
{0CCE9224-69AE-11D9-BED3-505054503030}	Identifies the File Share audit subcategory. This subcategory audits attempts to access a shared folder.
{0CCE9222-69AE-11D9-BED3-505054503030}	Identifies the Application Generated audit subcategory. This subcategory audits applications that generate events by using the Windows Auditing application programming interfaces (APIs). Applications designed to use the Windows Auditing API use this subcategory to log auditing events related to their function.
{0CCE9221-69AE-11D9-BED3-505054503030}	Identifies the Certification Services audit subcategory. This subcategory audits Active Directory Certificate Services (AD CS) operations.
{0CCE9220-69AE-11D9-BED3-505054503030}	Identifies the SAM audit subcategory. This subcategory audits events generated by attempts to access Security Accounts Manager (SAM) objects.
{0CCE921F-69AE-11D9-BED3-505054503030}	Identifies the Kernel Object audit subcategory. This subcategory audits attempts to access the system kernel, which include mutexes and semaphores. Only kernel objects with a matching SACL generate security audit events. Note: The Audit: Audit the access of global system objects policy setting controls the default SACL of kernel objects.
{0CCE921E-69AE-11D9-BED3-505054503030}	Identifies the Registry audit subcategory. This subcategory audits attempts to access registry objects. A security audit event is generated only for objects that have SACLs and only if the type of access requested, such as Read, Write, or Modify, and the account making the request match the settings in the SACL.
{0CCE921D-69AE-11D9-BED3-505054503030}	Identifies the File System audit subcategory. This subcategory audits user attempts to access file system objects. A security audit event is generated only for objects that have SACLs and only if the type of access requested, such as Write, Read, or Modify, and the account making the request match the settings in the SACL.
{0CCE9229-69AE-11D9-BED3-505054503030}	Identifies the Non Sensitive Privilege Use audit subcategory. This subcategory audits events generated by the use of nonsensitive privileges (user rights), such as logging on locally or with a Remote Desktop connection, changing the system time, or removing a computer from a docking station.

SubcategoryGUID	Purpose
{0CCE922A-69AE-11D9-BED3-505054503030}	Identifies the Other Privilege Use Events audit subcategory.
{0CCE9228-69AE-11D9-BED3-505054503030}	Identifies the Sensitive Privilege Use audit subcategory. This subcategory audits events generated by the use of sensitive privileges (user rights), such as acting as part of the operating system, backing up files and directories, impersonating a client computer, or generating security audits.
{0CCE922D-69AE-11D9-BED3-505054503030}	Identifies the DPAPI Activity audit subcategory. This subcategory audits events generated when encryption or decryption requests are made to the Data Protection application interface (DPAPI). DPAPI is used to protect secret information such as stored password and key information.
{0CCE922C-69AE-11D9-BED3-505054503030}	Identifies the Process Termination audit subcategory. This subcategory audits events generated when a process ends.
{0CCE922B-69AE-11D9-BED3-505054503030}	Identifies the Process Creation audit subcategory. This subcategory audits events generated when a process is created or starts. The name of the application or user that created the process is also audited.
{0CCE922E-69AE-11D9-BED3-505054503030}	Identifies the RPC Events audit subcategory. This subcategory audits inbound remote procedure call (RPC) connections.
{0CCE9232-69AE-11D9-BED3-505054503030}	Identifies the MPSSVC Rule-Level Policy Change audit subcategory. This subcategory audits events generated by changes in policy rules used by Windows Firewall.
{0CCE9234-69AE-11D9-BED3-505054503030}	Identifies the Other Policy Change Events audit subcategory. This subcategory audits events generated by other security policy changes that are not audited in the Policy Change category.
{0CCE9233-69AE-11D9-BED3-505054503030}	Identifies the Filtering Platform Policy Change audit subcategory. This subcategory audits events generated by changes to Windows Filtering Platform (WFP).
{0CCE922F-69AE-11D9-BED3-505054503030}	Identifies the Audit Policy Change audit subcategory. This subcategory audits changes in security audit policy settings.
{0CCE9231-69AE-11D9-BED3-505054503030}	Identifies the Authorization Policy Change audit subcategory. This subcategory audits events generated by changes to the authorization policy.
{0CCE9230-69AE-11D9-BED3-505054503030}	Identifies the Authentication Policy Change audit subcategory. This subcategory audits events generated by changes to the authentication policy.
{0CCE923A-69AE-11D9-BED3-505054503030}	Identifies the Other Account Management Events audit subcategory. This subcategory audits events generated by other user account changes that are not covered in this category.
{0CCE9239-69AE-11D9-BED3-505054503030}	Identifies the Application Group Management audit subcategory. This subcategory audits events generated by changes to application groups.
{0CCE9238-69AE-11D9-BED3-505054503030}	Identifies the Distribution Group Management audit subcategory. This subcategory audits events generated by changes to distribution groups.
{0CCE9237-69AE-11D9-BED3-505054503030}	Identifies the Security Group Management audit subcategory. This subcategory audits events generated by changes to security groups.
{0CCE9236-69AE-11D9-BED3-505054503030}	Identifies the Computer Account Management audit subcategory.

SubcategoryGUID	Purpose
BED3-505054503030}	This subcategory audits events generated by changes to computer accounts, such as when a computer account is created, changed, or deleted.
{0CCE9235-69AE-11D9-BED3-505054503030}	Identifies the User Account Management audit subcategory. This subcategory audits changes to user accounts.
{0CCE923E-69AE-11D9-BED3-505054503030}	Identifies the Detailed Directory Service Replication audit subcategory. This subcategory audits events generated by detailed AD DS replication between domain controllers (DCs) .
{0CCE923B-69AE-11D9-BED3-505054503030}	Identifies the Directory Service Access audit subcategory. This subcategory audits events generated when an AD DS object is accessed. Only AD DS objects with a matching SACL are logged.
{0CCE923D-69AE-11D9-BED3-505054503030}	Identifies the Directory Service Replication audit subcategory. This subcategory audits replication between two AD DS DCs.
{0CCE923C-69AE-11D9-BED3-505054503030}	Identifies the Directory Service Changes audit subcategory. This subcategory audits events generated by changes to AD DS objects. Events are logged when an object is created, deleted, modified, moved, or undeleted.
{0CCE9241-69AE-11D9-BED3-505054503030}	Identifies the Other Account Logon Events audit subcategory. This subcategory audits events generated by responses to credential requests submitted for a user account logon that are not credential validation or Kerberos tickets.
{0CCE9240-69AE-11D9-BED3-505054503030}	Identifies the Kerberos Service Ticket Operations audit subcategory. This subcategory audits events generated by Kerberos service ticket requests.
{0CCE923F-69AE-11D9-BED3-505054503030}	Identifies the Credential Validation audit subcategory. This subcategory audits events generated by validation tests on user account logon credentials.
{0CCE9242-69AE-11D9-BED3-505054503030}	Identifies the Kerberos Authentication Service audit subcategory. This subcategory audits events generated by Kerberos authentication ticket-granting ticket (TGT) requests.
{0CCE9245-69AE-11D9-BED3-505054503030}	Identifies the Removable Storage audit subcategory. This subcategory audits user attempts to access file system objects on any Removable Storage device. A security audit event is generated for every read or write access to a file object on any Removable Storage device attached to the user's machine.
{0CCE9246-69AE-11D9-BED3-505054503030}	Identifies the Central Access Policy Staging audit subcategory. This subcategory audits access requests where the permission granted or denied by a proposed policy differs from that granted or denied by the current central access policy on an object.
{0CCE9247-69AE-11D9-BED3-505054503030}	Identifies the User/Device Claims audit subcategory. This subcategory SHOULD <3> audit the user and device claims that are present in the token of an associated logon.
{0CCE9248-69AE-11D9-BED3-505054503030}	Identifies the PNP Activity audit subcategory. This subcategory SHOULD <4> audit events generated by plug and play (PNP).
{0CCE9249-69AE-11D9-BED3-505054503030}	Identifies the Group Membership audit subcategory. This subcategory SHOULD <5> audit the group membership of a token for an associated logon.

2.2.1.3 Inclusion Setting, Exclusion Setting, and Setting Value

This section defines settings that enable an administrator to define whether to add or remove a subcategory from the client advanced audit policy.

The possible value of these attributes depends whether the subcategory audit setting policy target is "System" or a specific user or group.

2.2.1.3.1 Inclusion Setting, Exclusion Setting, and SettingValue for System Audit Subcategories

This section defines the syntax for the InclusionSetting, ExclusionSetting, and SettingValue attributes when the PolicyTarget attribute is set to "System".

The syntax for the entries in this category MUST be as follows.

```
InclusionSetting-SA = "Success" / "Failure" / "Success and Failure" / "No Auditing" / "Not Specified"  
ExclusionSetting-SA = ""  
SettingValue-SA = 1*DIGIT
```

Please note that the element names above have a postfix of "-SA" to differentiate them from per-user audit settings which have a postfix of "-UA".

The value of SettingValue MUST be one of the following:

3. A value of "0": Indicates that this audit subcategory setting is unchanged.
4. A value of "1": Indicates that this audit subcategory setting is set to Success Audits Only.
5. A value of "2": Indicates that this audit subcategory setting is set to Failure Audits Only.
6. A value of "3": Indicates that this audit subcategory setting is set to Success and Failure Audits.
7. A value of "4": Indicates that this audit subcategory setting is set to None.

Note The value of InclusionSetting is for user readability only and is ignored when the advanced audit policy is applied by the audit configuration client-side plug-in.

2.2.1.3.2 Inclusion Setting, Exclusion Setting, and SettingValue for Per-User Audit Subcategories

This section defines the syntax for the InclusionSetting, ExclusionSetting, and SettingValue attributes when the PolicyTarget attribute is set to a specific user or group SID.

The syntax for the entries in this category MUST be as follows.

```
InclusionSetting-UA = "SettingValueText"  
ExclusionSetting-UA = SettingValueText  
SettingValueText-UA = "Success" / "Failure" / "Success and Failure" / "No Auditing" / "Not Specified"  
  
SettingValue-UA = 1*DIGIT
```

Note that the element names above have a postfix of "-UA" to differentiate them from System advanced audit policy settings, which have a postfix of "-SA".

The attribute **SettingValueText** is for user readability only and is ignored when the advanced audit policy is applied by the audit configuration client-side plug-in.

The value of SettingValue MUST be one of the following:

- A value of "0": Indicates that this audit subcategory setting is unchanged.
- A value of "16": Indicates that this audit subcategory setting is set to None.
- A decimal numerical value created by combining the following bits.

Bit order	Hexadecimal value	Purpose
1	0x01	Include Success: This bit will cause a Success Audit to be generated even if not specified by the system advanced audit policy.
2	0x02	Exclude Success: This bit will cause a Success Audit to be suppressed regardless of the system advanced audit policy. This setting is not honored for users who are members of the Administrators local group.
3	0x04	Include Failure: This bit will cause a Failure Audit to be generated even if not specified by the system advanced audit policy.
4	0x08	Exclude Failure: This bit will cause a Failure Audit to be suppressed regardless of the system advanced audit policy. This setting is not honored for users who are members of the Administrators local group.

Note Include has a higher precedence than exclude:

- If **Include Success** and **Exclude Success** bits are set, **Include Success** is used and Exclude Success is ignored.
- If **Include Failure** and **Exclude Failure** bits are set, **Include Failure** is used and Exclude Failure is ignored.[<6>](#)

2.2.2 Audit Options

This section defines settings that enable an administrator to set the audit options for an advanced audit policy. The syntax for the entries in this category MUST be as follows.

```
AuditOptions = MachineName ",,Option:" AuditOptionType ",," AuditOptionValueText ",,"  
AuditOptionValue
```

2.2.2.1 Audit Option Type

This section defines the advanced audit options that are part of the audit policy. The syntax for the entries in this category MUST be as follows.

```
AuditOptionType = String
```

The value of AuditOptionType MUST be one of the following:

AuditOptionType	Purpose
CrashOnAuditFail	This audit option specifies whether the client shuts down if it is unable to log security events. If this security setting is enabled, it causes the client to stop if a security audit cannot be logged for any reason.
FullPrivilegeAuditing	This audit option specifies whether the client generates an event when one or more of these privileges are assigned to a user security token: 1. AssignPrimaryTokenPrivilege 2. AuditPrivilege 3. BackupPrivilege 4. CreateTokenPrivilege 5. DebugPrivilege 6. EnableDelegationPrivilege 7. ImpersonatePrivilege 8. LoadDriverPrivilege 9. RestorePrivilege 10. SecurityPrivilege 11. SystemEnvironmentPrivilege 12. TakeOwnershipPrivilege 13. TcbPrivilege
AuditBaseObjects	This security setting specifies whether to audit the access of global system objects. If this audit option is enabled, it causes system objects, such as mutexes, events, semaphores, and DOS devices, to be created with a default system access control list (SACL) . Only named objects are given a SACLs; SACL are not given to objects without names. If the Kernel Object audit subcategory is also enabled, access to these system objects is audited.
AuditBaseDirectories	The AuditBaseDirectories option specifies that named kernel objects (such as mutexes and semaphores) are to be given SACLs when they are created. AuditBaseDirectories affects container objects while AuditBaseObjects affects objects that cannot contain other objects.

2.2.2.2 Audit Option Value

This section defines the possible values corresponding to the audit options. The syntax for the entries in this category **MUST** be as follows.

```
AuditOptionValueText = "Enabled" / "Disabled"
AuditOptionValue = 1 * DIGIT
```

Note The **AuditOptionValueText** field is for user reference only and is ignored when the advanced audit policy is applied by the audit configuration client-side plug-in.

The value of AuditOptionValue MUST be one of the following:

AuditOptionValue	Purpose
"0"	The audit option is disabled.
"1"	The audit option is enabled.

2.2.3 Global Object Access Audit Settings

This section defines settings that enable an administrator to set the global object access auditing settings for an advanced audit policy.

Global object access audit settings are used by administrators to define system access control lists (SACLs) that apply dynamically to every object in a specific resource manager. When a global object access audit setting is defined, the auditing system combines the SACL defined in the security descriptor that is being accessed with the global object access SACL for the corresponding resource manager. An event is logged if either of the two SACLs (object SACL or global SACL) determines that the activity is to be audited.

The syntax for the entries in this category MUST be as follows.

```
GlobalObjectAccessAuditSettings = MachineName ".,," ResourceGlobalSaclType ".,.,," GlobalSACL
```

2.2.3.1 Resource Global SACL Type

This section defines the use of the ResourceGlobalSaclType attribute. The syntax for the entries in this category MUST be as follows.

```
ResourceGlobalSaclType = "FileGlobalSacl" / "RegistryGlobalSacl"
```

The value of ResourceGlobalSaclType MUST be one of the following:

ResourceGlobalSaclType	Purpose
"FileGlobalSacl"	Defines a global SACL for the File System resource manager.
"RegistryGlobalSacl"	Defines a global SACL for the Registry resource manager.

2.2.3.2 Global System Access Control List (SACL)

This section defines the use of the GlobalSACL attribute. The syntax for the entries in this category MUST be as follows.

```
GlobalSACL = SDDLString  
SDDLString = String
```

The GlobalSACL attribute MUST be in the form of an SDDL encoding of a SACL of a security descriptor. For more information, see [\[MSDN-SDDL\]](#).

2.2.4 Machine Name

This section defines the use of the machine name, used on different sections of the advanced audit policy. The syntax for the entries in this category MUST be as follows.

```
MachineName = String / QuotedString
```

The machine name is given for user reference and is ignored when the audit configuration client-side plug-in applies an advanced audit policy.

3 Protocol Details

3.1 Audit Configuration Protocol Administrative-Side Plug-in Details

The audit configuration protocol administrative-side plug-in participates in the advanced audit policy authoring and assignment steps, as specified in section [2](#). The advanced audit policy MUST be stored as a text file by using a .csv format, as specified in section [2.2](#). The advanced audit policies MUST be stored in a location accessible by using file access protocol sequences.

3.1.1 Abstract Data Model

The audit configuration protocol administrative-side plug-in maintains no state. It loads all the settings, as specified in section [2.2](#), in memory.

The administrative-side plug-in is used, through the implementation-specific tool providing a graphical user interface, to interact with the advanced audit policy file, as specified in [\[MS-GPOL\]](#). The plug-in determines the physical location of a desired policy, creates a new policy, or opens an existing policy as appropriate, and displays it to the administrator. After the administrator modifies the policy, the changes MUST be propagated back into the policy at the specified location.

3.1.2 Timers

None.

3.1.3 Initialization

The process for reading the settings from the GPO for administrative purposes MUST be the same as those as specified in section [3.2.5](#), steps 1-3.

3.1.4 Higher-Layer Triggered Events

The administrative-side plug-in is triggered when an administrator starts an **administrative tool**. The plug-in displays the current settings to the administrator, and when the administrator requests a change in settings, the plug-in updates the stored configuration appropriately as specified in section [2.2](#).

For both viewing and editing settings, the administrative-side plug-in MUST first open the specified GPO to fetch its network path. The plug-in MUST attempt to read an audit.csv file with the settings from the following location (for viewing) or write to the following location (for editing): <gpo path>\Microsoft\Windows NT\Audit\audit.csv (where <gpo path> is the **computer-scoped Group Policy Object path**, if the computer settings are being viewed or updated).

File reads and writes MUST be performed, as specified in [\[MS-GPOL\]](#) section 3.3. File names and paths SHOULD be regarded as case-insensitive. If the copy fails, the administrative-side plug-in MUST display to the user that the operation failed.

3.1.5 Message Processing Events and Sequencing Rules

The administrative-side plug-in reads extension-specific data from the remote storage location and passes that information to an implementation-specific tool that provides a graphical user interface to display the current settings to an administrator.

The administrative-side plug-in creates the advanced audit policy file in the remote location specified in section [3.1.4](#) if the file does not exist. The administrative-side plug-in writes the extension-specific

configuration data to the remote storage location if the administrator makes any changes to the existing configuration.

After every creation, modification, or deletion that affects an audit policy file on SYSVOL, the administrative-side plug-in MUST invoke the Group Policy Extension Update task, as specified in [\[MS-GPOL\]](#) section 3.3.4.4.

3.1.6 Timer Events

None.

3.1.7 Other Local Events

None.

3.2 Advanced Audit Policy Configuration Client-Side Plug-in Details

The advanced audit policy configuration client-side plug-in interacts with the Group Policy framework, as specified in [\[MS-GPOL\]](#) section 3.2. This plug-in MUST receive the advanced audit policy and modify the appropriate part of the Abstract Data Model (ADM) for each element in the policy as specified in this section.

3.2.1 Abstract Data Model

This section defines a conceptual model of possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to explain how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with what is described in this document.

3.2.1.1 Policy Setting State

The client-side plug-in persistent state consists of the 4 sections below. The location where this state is stored is specific to each implementation.

- System Advanced Audit Policy:
 - A list of records, each with a record identifier (subcategory GUID). In addition to the key, each record has an audit setting value that defines the audit behavior for the subcategory. Subcategory settings are specified in section [2.2.1](#).
- Per-User Advanced Audit Policy:
 - A list of records, each with a user or group SID, a subcategory GUID, and an audit setting value that overrides the system audit behavior for the subcategory for the user or group.
- Audit Options:
 - A list of records, each with an audit option type and a setting value. Audit options are specified in section [2.2.2](#).
- Global Object Access Auditing:
 - Consists of two persistent **SACL**-valued data elements: FileGlobalSacl and RegistryGlobalSacl. This is used to store the global object access audit settings used by administrators to define system access control lists (SACLs) that apply dynamically to every object in a specific resource manager. For more details, see section [2.2.3](#).

3.2.2 Timers

None.

3.2.3 Initialization

When invoked by the Group Policy framework with a list of one or more applicable GPOs, the audit configuration protocol client-side plug-in MUST do the following: locate all the advanced audit policy files within those GPOs, copy the policies to the local machine, read the policies, and apply them as specified in section [3.2.5](#).

Locating advanced audit policy files MUST be done by using the Group Policy: Core Protocol, as specified in [\[MS-GPOL\]](#) section 3.2.5.1, and the LDAP search protocol, as specified in [\[RFC2251\]](#) section 4.5. The policy files MUST be copied and read by using file access protocol sequences.

3.2.4 Higher-Layer Triggered Events

This plug-in implements one higher-layer triggered event: [Process Group Policy](#).

3.2.4.1 Process Group Policy

The plug-in implements the Process Group Policy abstract event interface, as specified in [\[MS-GPOL\]](#) section 3.2.4.1. The plug-in does not make use of the *Deleted GPOs*, the flags, or the security tokens arguments. When the event is triggered, the audit configuration protocol client-side plug-in MUST take the actions described in the section [3.2.5](#).

3.2.5 Message Processing Events and Sequencing Rules

The audit configuration protocol client-side extension MUST be invoked by the Group Policy framework whenever applicable GPOs need to be processed, as specified in [\[MS-GPOL\]](#) section 3.2.5.1. On such an event, the audit configuration protocol client-side plug-in MUST take the actions described in this section.

When invoked, the audit configuration protocol client-side plug-in expects a list of applicable GPOs in the "New or changed GPOs" parameter. It MUST then go through this list and, for each GPO, locate and retrieve the contained advanced audit policy. For each of those GPOs, one file with the format (as specified in section [2.2](#)) MUST be copied from the Group Policy: Core Protocol server. If any file cannot be read, the plug-in MUST log information about the failure and continue to copy files for other GPOs.

For each GPO, the advanced audit policy configuration client-side plug-in MUST generate the following file access sequences to copy the file locally:

Sequence	Description
File Open from Client to Server	The plug-in MUST attempt to open the file specified by <scoped gpo path>\Microsoft\Windows NT\Audit\audit.csv.
File Read Sequences	One or more file reads MUST be done to read the entire content of the opened file or until an error occurs.
File Close	A file close operation MUST be performed.

The file MUST be parsed according to the format specified in section 2.2. If the file does not conform to that format, the entire configuration operation MUST be ignored. If the file does conform to that format, the settings MUST be applied to the corresponding audit parameters on the system.

After all the advanced audit policies are retrieved, each policy MUST be opened and the contained advanced audit policy settings MUST be extracted and applied for each ADM element corresponding to section 2.2.

When reading the advanced audit policy configuration file, the client-side extension follows the logical flow mentioned below.

If the "Policy Target" column value is empty AND if the "Subcategory" column value indicates FileGlobalSacl, process the "Setting Value" column value in the following way:

- Convert the "Setting Value" column value into a security descriptor based on the format defined in [\[MSDN-SDDL\]](#).
- For each Audit Access Control Entry (ACE) in the SAcl of the security descriptor extracted in the previous step, add it to the **FileGlobalSacl** ADM variable if it doesn't already exist.

If the "Policy Target" column value is empty AND if the "Subcategory" column value indicates RegistryGlobalSacl, process the "Setting Value" column value in the following way:

- Convert the "Setting Value" column value into a security descriptor based on the format defined in [\[MSDN-SDDL\]](#).
- For each Audit Access Control Entry (ACE) in the SAcl of the security descriptor extracted in the previous step, add it to the **RegistryGlobalSacl** ADM variable if it doesn't already exist. [<7>](#)

If the "Policy Target" column value is empty, then verify that the "Subcategory" column value is one of those specified in section [2.2.2.1](#), Audit Option Type. Once verified, store the "Setting Value" column value in the **AuditOptionValue** field of the corresponding AuditOptionType in the **Audit Options** ADM variable as specified in section [3.2.1.1](#).

If the "Exclusion Setting" column value is empty, then verify that the "Subcategory GUID" column value is one of those specified in [Subcategory and SubcategoryGUID \(section 2.2.1.2\)](#). Once verified, store the "Setting Value" column value in the audit setting value field of the corresponding subcategory GUID in the **System Advanced Audit Policy** ADM variable as specified in section 3.2.1.1.

If both the "Policy Target" and the "Exclusion Setting" column values are not empty, then verify that the "Subcategory GUID" column value is one of those specified in section Subcategory and SubcategoryGUID (section 2.2.1.2). Once verified, for the user identified by the "Policy Target" column value, store the "Setting Value" column value in the audit setting value field of the corresponding subcategory GUID in the **Per-User Advanced Audit Policy** ADM variable as specified in section 3.2.1.1.

3.2.6 Timer Events

None.

3.2.7 Other Local Events

None.

4 Protocol Examples

4.1 Example Involving System Audit Subcategory Settings

In the following example, an administrator specifies that the designated audit settings be applied for computers to which a certain GPO applies:

- Exclude audit attempts for IPsec Driver.
- Audit successful attempts for System Integrity.
- Audit successful and failed attempts for IPsec Extended Mode.
- Leave the File System policy unchanged.

```
Machine Name,Policy Target,Subcategory,Subcategory GUID,Inclusion Setting,Exclusion  
Setting,Setting Value  
TEST-MACHINE,System,IPsec Driver,{0CCE9213-69AE-11D9-BED3-505054503030},No Auditing,,0  
TEST-MACHINE,System,System Integrity,{0CCE9212-69AE-11D9-BED3-505054503030},Success,,1  
TEST-MACHINE,System,IPsec Extended Mode,{0CCE921A-69AE-11D9-BED3-505054503030},Success and  
Failure,,3  
TEST-MACHINE,System,File System,{0CCE921D-69AE-11D9-BED3-505054503030},Not specified,,0
```

4.2 Example Involving Per-User Audit Subcategory Settings

In the following example, an administrator specifies that the designated audit settings be applied for computers to which a certain GPO applies:

- Include made successful attempts for File System for user S-1-5-21-2127521184-1604012920-1887927527-123456.
- Exclude made failed attempts for File System for user S-1-5-21-2127521184-1604012920-1887927527-123456.

```
Machine Name,Policy Target,Subcategory,Subcategory GUID,Inclusion Setting,Exclusion  
Setting,Setting Value  
TEST-MACHINE,S-1-5-21-2127521184-1604012920-1887927527-123456,File System,{0CCE921D-69AE-  
11D9-BED3-505054503030},Success,Failure,9
```

4.3 Example Involving Audit Options

In the following example, an administrator specifies that the designated audit settings be applied for computers to which a certain GPO applies:

- Enable audit option CrashOnAuditFail.
- Disable audit option FullPrivilegeAuditing.
- Disable audit option AuditBaseObjects.
- Disable audit option AuditBaseDirectories.

```
Machine Name,Policy Target,Subcategory,Subcategory GUID,Inclusion Setting,Exclusion  
Setting,Setting Value
```

```

TEST-MACHINE,,Option:CrashOnAuditFail,,Enabled,,1
TEST-MACHINE,,Option:FullPrivilegeAuditing,,Disabled,,0
TEST-MACHINE,,Option:AuditBaseObjects,,Disabled,,0
TEST-MACHINE,,Option:AuditBaseDirectories,,Disabled,,0

```

4.4 Example Involving Global Object Access Auditing

In the following example, an administrator specifies that the designated audit settings be applied for computers to which a certain GPO applies:

1. Set a registry Global SACL to log all the activity for everyone.

```

Machine Name,Policy Target,Subcategory,Subcategory GUID,Inclusion Setting,Exclusion
Setting,Setting Value
TEST-MACHINE,,RegistryGlobalSacl,,,,S: (AU;SA;FA;;;WD)

```

4.5 Example of Configuring Multiple Types of Settings

In the following example, an administrator specifies that for computers to which a certain GPO applies, all the settings specified in the previous sections are configured as designated.

```

Machine Name,Policy Target,Subcategory,Subcategory GUID,Inclusion Setting,Exclusion
Setting,Setting Value
TEST-MACHINE,System,IPsec Driver,{0CCE9213-69AE-11D9-BED3-505054503030},No Auditing,,0
TEST-MACHINE,System,System Integrity,{0CCE9212-69AE-11D9-BED3-505054503030},Success,,1
TEST-MACHINE,System,IPsec Extended Mode,{0CCE921A-69AE-11D9-BED3-505054503030},Success and
Failure,,3
TEST-MACHINE,System,File System,{0CCE921D-69AE-11D9-BED3-505054503030},Not specified,,0
TEST-MACHINE,S-1-5-21-2127521184-1604012920-1887927527-123456,File System,{0CCE921D-69AE-
11D9-BED3-505054503030},Success,Failure,9
TEST-MACHINE,,Option:CrashOnAuditFail,,Enabled,,1
TEST-MACHINE,,Option:FullPrivilegeAuditing,,Disabled,,0
TEST-MACHINE,,Option:AuditBaseObjects,,Disabled,,0
TEST-MACHINE,,Option:AuditBaseDirectories,,Disabled,,0
TEST-MACHINE,,RegistryGlobalSacl,,,,S: (AU;SA;FA;;;WD)

```

5 Security

5.1 Security Considerations for Implementers

Setting both the advanced audit policies (as described in this document) and the event audit policies (as described in [\[MS-GPSB\]](#) section 2.2.4) on the same client can lead to inconsistent behavior. Therefore, if the advanced audit policies are used on a client, it is recommended that the registry value `MACHINE\System\CurrentControlSet\Control\LSA\SCENoApplyLegacyAuditPolicy` be set to 1, using the mechanism described in [\[MS-GPSB\]](#) section 2.2.5. This avoids the conflict by preventing the event audit policies from being applied on the client.

5.2 Index of Security Parameters

5.2.1 Security Parameters Affecting Behavior of the Protocol

Security Parameter	Explanation of setting
<code>MaxNoGPOListChangesInterval</code> [MS-GPOL] section 3.2.1.24	When the value of the <code>MaxNoGPOListChangesInterval</code> for a particular client-side extension is set (by local implementation-specific means) to a nonzero integer value, the Group Policy client will invoke the extension after <code>MaxNoGPOListChangesInterval</code> minutes, even if the policy has not changed since the last invocation of the extension. This setting ensures that the advanced audit policy settings created by the administrator of a domain are reapplied on the client after <code>MaxNoGPOListChangesInterval</code> minutes. This limits the amount of time that the local and central advanced audit policy settings could be out of sync because of local modifications to the policy. <8>

5.2.2 System Security Parameters Carried by the Protocol

Settings category	Comments
Subcategory settings	For more information, see section 2.2.1 .
Audit options	For more information, see section 2.2.2 .
Global object access audit settings	For more information, see section 2.2.3 .

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows 7 operating system
- Windows Server 2008 R2 operating system
- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system
- Windows 10 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows 11 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 2.2.1.1](#): In Windows, audit settings associated with group **SID** strings are ignored by the client.

[<2> Section 2.2.1.2](#): In Windows, this subcategory also audits the following events:

- Startup and shutdown of the Windows Firewall.
- Security policy processing by the Windows Firewall.

[<3> Section 2.2.1.2](#): User/Device Claims audit subcategory is not implemented in Windows 7, Windows Server 2008 R2, Windows 8, Windows Server 2012, Windows 8.1, and Windows Server 2012 R2.

[<4> Section 2.2.1.2](#): PNP Activity is not implemented in Windows 7, Windows Server 2008 R2, Windows 8, Windows Server 2012, Windows 8.1, and Windows Server 2012 R2.

[<5> Section 2.2.1.2](#): Group Membership is not implemented in Windows 7, Windows Server 2008 R2, Windows 8, Windows Server 2012, Windows 8.1, and Windows Server 2012 R2.

<6> [Section 2.2.1.3.2](#): If any subcategory in the Per-User Advanced Audit Policy section is defined for a given user or group in Windows, the value **Include Failure** (0x4) is used as default for all the rest of the audit subcategories that are not defined for that user after all the applicable policies are processed. The **Include Failure** setting will cause a **Failure** Audit to be generated even if not specified by the system advanced audit policy.

<7> [Section 3.2.5](#): In Windows 7 and Windows Server 2008 R2, individual Audit ACEs from different GPOs are not merged into a single SACL; instead the final value of the FileGlobalSacl, as well as the RegistryGlobalSacl ADM variables, come from the GPO with the highest precedence (as described in [\[MS-GPOL\]](#)) where the setting is defined.

<8> [Section 5.2.1](#): In Windows, the value of MaxNoGPOListChangesInterval is 0x3c0 (960 minutes) for the advanced audit policy client-side extension.

7 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
6 Appendix A: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

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